

EDUCATION

M.Tech	Malaviya National Institute of Technology, Jaipur	2019 - 2021	8.75
	Computer Science and Engineering		Gold Medalist
B.Tech	Dr. Babasaheb Ambedkar Technological University, Lonere	2014 - 2018	8.9
	Computer Engineering		

SKILLS

Languages	C++ with STL, Python, Java
Cloud and Containerization	Azure, AWS, Docker, Kubernetes
Infrastructure as Code	Ansible, Terraform
Databases	MySQL, MongoDB
Tools and Technologies	Gitlab, Docker, Okta API, New Relic, GraphQL API, RESTful API
ML Libraries	PyTorch, Tensorflow, Keras, Numpy, OpenCV, Scikit-learn, Pandas
GenAI & Agentic Systems:	Model Context Protocol (MCP), LLM Agentic Workflows & Playbooks, Retrieval-Augmented Generation (RAG), Prompt Engineering & Structured Outputs.

TECHNICAL EXPERIENCE

Software Engineer — Embedded Systems Runtime and Firmware Nov 2024 — Present
Google Bangalore, India

- Pixel TPU & DSP Runtime Silicon Bring-Up: Drove EdgeTPU and DSP enablement for next-generation Google Tensor mobile SoCs, resolving early-silicon dynamic linker faults, platform detection logic, and kernel security permissions to achieve stable on-device ML execution.
- Distributed Bug Triage Platform: Implemented an automated bug triage service from the ground up using Google Cloud Spanner distributed locking to eliminate race conditions across microservices; reduced manual issue duplication by 50%, automated crash artifact/ramdump processing, and saved 5—10 engineering hours/week. Designed signature-matching rule engines in Python to parse logs and extract tombstones for instant triage context; co-designed a next-generation GenAI triage agent using LLM playbooks and Model Context Protocol (MCP).
- Platform-Wide NNAPI Deprecation: Authored technical justification and executed platform-level decommissioning of legacy NNAPI vendor HALs across Android; verified OS stability and quantified a 15.2x latency surge and 81% CPU thread stall across top 1P/3P applications to accelerate ecosystem transition to LiteRT.
- Kernel Driver Hang Detection: Designed an atomic RAIL tracker with 100% coverage over DSP kernel driver interactions, integrating with watchdog heartbeats to provide instantaneous root-cause visibility during driver deadlocks and hardware stalls.
- Memory & CPU Telemetry Frameworks: Developed automated telemetry tooling synchronizing userspace memory (showmap RSS/PSS), kernel DMA buffers (dmabuf_dump), and CPU cycles (simpleperf), identifying memory leaks and MIPS regressions across real-time Pixel camera ML pipelines.
- DSP Memory Architecture Optimization: Implemented C-API controls for dynamic memory pool sizing, buffer eviction mechanisms, and automated tightly-coupled memory (TCM) allocator resets during core initialization.
- Camera Automation & HW Verification: Built automated testing suites using Python and ADB covering all camera algorithms and use cases; developed low-level verification tests for critical hardware behaviors (DMA buffers, power management/DVFS), optimizing multi-firmware performance benchmarking.
- Dynamic Tensor Execution in LiteRT: Engineered support for dynamic tensor dimensions in LiteRT delegate kernels, enabling hardware-accelerated on-device execution for variable-input vision and language models.

Software Developer Sept 2022 — Nov 2024
Halliburton Development Center Bangalore, India

- Disaster Recovery Automation: Investigated and automated the failover process for Azure NetApp Files (ANF) volumes from primary to secondary during disaster recovery, including post-DR steps, to enhance system resilience and minimize recovery time.

- **Kubernetes Storage Migration:** Automated the migration and updating of **Kubernetes** Persistent Volume (PV) and Persistent Volume Claim (PVC) mount points from SoftNAS to ANF, ensuring seamless transitions with minimal downtime.
- **Cloud Secrets Management:** Implemented Landmark Secrets Management (LSM) to automate the secure rotation of credentials in **AWS Secrets Manager** and **Azure Key Vault**, enhancing cloud security practices.
- **Serverless Monitoring & Alerts:** Developed a **Python**-based error notification system using **NerdGraph GraphQL API**, deployed through **AWS Lambda** and **Azure Functions** via **Terraform** for automated monitoring.
- **Customer Onboarding Automation:** Developed an automated customer on-boarding system by integrating Okta API with Python and Bash scripting, significantly reducing the workload on the support team by **80%**.
- **CI/CD & Cloud Infrastructure:** Strong experience in **IAC**, **CI/CD**, **REST APIs**, and working within **Agile** and cross-functional teams.

Team Lead DS365.ai Support, Cloud Support Analyst II
Halliburton Development Center

Aug 2021 — Sept 2022
Bangalore, India

- **Platform Support Leadership:** Led a dynamic support team for the "DS365.ai" enterprise platform, ensuring high-quality customer interactions, fast SLA resolution, and critical incident escalation.
- **Support Optimization & Knowledge Base:** Developed and implemented new support strategies, including the creation of comprehensive knowledge base articles, reducing recurring inquiries by **30%**.

PUBLICATIONS & RESEARCH

MuSTAT: Face Ageing using Multi-Scale Target Age Style Transfer
Best Paper Award | *Communications in Computer and Information Science, Springer*

CVIP 2023
[\[Springer Link\]](#)

- **Master's Thesis Outcome:** Published and presented as first-rate research at the 8th International Conference on Computer Vision and Image Processing (CVIP 2023, IIT Jammu), winning the **Best Paper Award**.
- Proposed a novel generative encoder-decoder architecture with **Selective Transfer Units (STUs)** across skip connections, adaptively modulating identity features with target age style distributions.
- Demonstrated superior continuous age progression/regression and verified identity consistency across **6 Face Recognition Systems (89.82% FNMR @ 0.0001% FMR)**.